

# Problem 1:

Kruskal

edges:

$$AB = 2$$

$$CD = 3$$

$$FE = 4$$

$$CG = 5$$

$$GD = 6$$

$$GE = 7$$

$$AF = 8$$

$$FB = 9$$

$$BG = 10$$

$$BC = 11$$

$$ED = 12$$

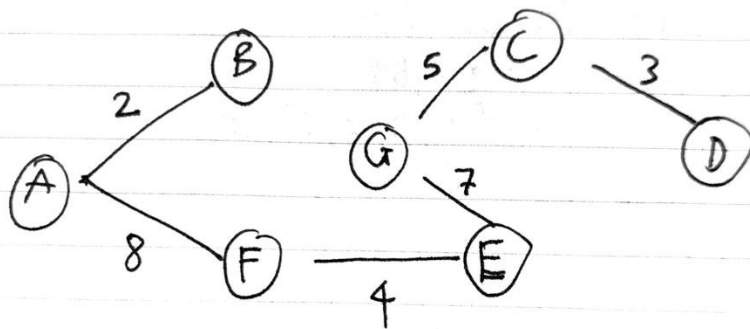
$$FG = 13$$

| edge | decision | MST                      |
|------|----------|--------------------------|
| AB   | take     | {AB}                     |
| CD   | take     | {AB, CD}                 |
| FE   | take     | {AB, CD, FE}             |
| CG   | take     | {AB, CD, FE, CG}         |
| GD   | no       | {AB, CD, FE, CG}         |
| GE   | take     | {AB, CD, FE, CG, GE}     |
| AF   | take     | {AB, CD, FE, CG, GE, AF} |

We have  $|V| = 7 \Rightarrow$  MST must have  $|V| - 1$  edge, we stop here because we have 6 edges in MST

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we take source  $S = \{A\}$   
other vertices  $V = \{B, C, D, E, F, G\}$

- smallest edge from  $\{A\}$   
is AB, then  $S = \{A, B\}$

$V = \{C, D, E, F, G\}$

- smallest edge from  $\{A, B\}$

is AF, then  $S = \{A, B, F\}$

$V = \{C, D, E, G\}$

- and so we have FE then  $S = \{A, B, F, E\}$

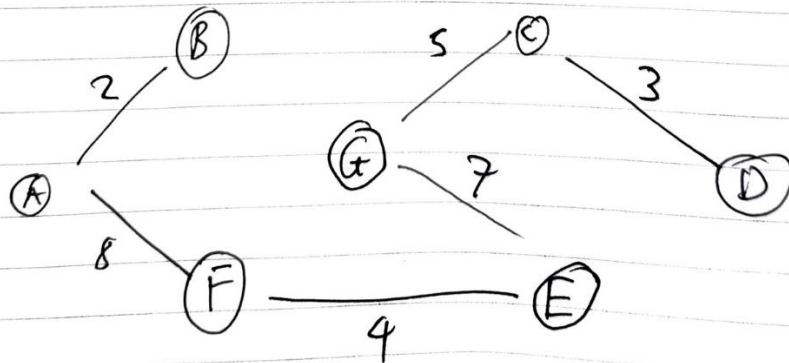
$V = \{C, D, G\}$

- then we have EG then  $S = \{A, B, F, E, G\}$

$V = \{C, D\}$

- then GC then  $S = \{A, B, F, E, G, C\}$

finally : CD then  $S = \{A, B, F, E, G, C, D\}$



## Problem 2:

2,

when we start, let consider path value, start at A

(1), (2), (3), (4), (5)

0  $\infty$   $\infty$   $\infty$   $\infty$

first loop

$$1 \rightarrow 2 = 6 < \infty \Rightarrow (2) = 6$$

$$1 \rightarrow 4 = 7 < \infty \Rightarrow (4) = 7$$

$$2 \rightarrow 3 = 6 + 5 = 11 < \infty \Rightarrow (3) = 11$$

$$2 \rightarrow 4 = 6 + 8 = 14 > (4) \Rightarrow \text{no change}$$

$$2 \rightarrow 5 = 6 - 4 = 2 < \infty \Rightarrow (5) = 2$$

$$3 \rightarrow 2 = 11 - 2 = 9 > (2) \Rightarrow \text{no change}$$

$$4 \rightarrow 3 = 7 - 3 = 4 < (3) \Rightarrow (3) = 4$$

$$4 \rightarrow 5 = 7 + 9 = 16 > (5) \Rightarrow \text{no change}$$

$$5 \rightarrow 1 = 2 + 2 = 4 > (1) \Rightarrow \text{no change}$$

$$5 \rightarrow 3 = 2 + 7 = 9 > (3) \Rightarrow \text{no change}$$

then after first loop:

(1), (2), (3), (4), (5)

0 6 4 7 2

2nd loop

$$1 \rightarrow 2 = 0 + 6 = 6 > (2) \Rightarrow \text{no}$$

$$1 \rightarrow 4 = 0 + 7 = 7 > (4) \Rightarrow \text{no}$$

$$2 \rightarrow 3 = 6 + 5 = 11 > (3) \Rightarrow \text{no}$$

$$2 \rightarrow 4 = 6 + 8 = 14 > (4) \Rightarrow \text{no}$$

$$2 \rightarrow 5 = 6 - 4 = 2 < (5) \Rightarrow \text{no}$$

$$3 \rightarrow 2 = 4 - 2 = 2 < (2) \Rightarrow (2) = 2$$

$$4 \rightarrow 3 = 7 - 3 = 4 > (3) \Rightarrow \text{no}$$

$$4 \rightarrow 5 = 7 + 9 = 16 > (5) \Rightarrow \text{no}$$

$$5 \rightarrow 1 = 2 + 2 = 4 > (1) \Rightarrow \text{no}$$

$$5 \rightarrow 3 = 2 + 7 = 9 > (3) \Rightarrow \text{no}$$

(1), (2), (3), (4), (5)

0 2 4 7 2

| 3rd    | 4th    |
|--------|--------|
| 1 : 0  | 1 : 0  |
| 2 : 2  | 2 : 2  |
| 3 : 4  | 3 : 4  |
| 4 : 7  | 4 : 7  |
| 5 : -2 | 5 : -2 |

### Problem 3:

|                                             |               |              |
|---------------------------------------------|---------------|--------------|
| 3, vertices: <del>A, B, C, D, E, F, G</del> |               |              |
| Start : A : 0                               | take A:       | A : 0        |
| B : $\infty$                                |               | B : $\infty$ |
| C : $\infty$                                |               | C : 1 A      |
| D : $\infty$                                |               | D : 2 A      |
| E : $\infty$                                |               | E : $\infty$ |
| F : $\infty$                                |               | F : $\infty$ |
| G : $\infty$                                |               | G : $\infty$ |
| take C: A : 0                               | take D:       | A : 0        |
| B : 3 C                                     |               | B : 3 C      |
| C : 1 A                                     |               | C : 1 A      |
| D : 2 A                                     |               | D : 2 A      |
| E : 4 C                                     |               | E : 4 C      |
| F : $\infty$                                |               | F : $\infty$ |
| G : $\infty$                                |               | G : 3 D      |
| take G: A : 0                               | take B: A : 0 |              |
| B : 3 C                                     |               | B : 3 C      |
| C : 1 A                                     |               | C : 1 A      |
| D : 2 A                                     |               | D : 2 A      |
| E : 4 C                                     |               | E : 4 C      |
| F : 4 G                                     |               | F : 4 G      |
| G : 3 D                                     |               | G : 3 D      |

|            |       |   |             |       |
|------------|-------|---|-------------|-------|
| choose E : | A : 0 |   | finally F : | A : 0 |
|            | B : 3 | C |             | B : 3 |
|            | C : 1 | A |             | C : 1 |
|            | D : 2 | A |             | D : 2 |
|            | E : 4 | C |             | E : 4 |
|            | F : 4 | G |             | F : 4 |
|            | G : 3 | D |             | G : 3 |

stop here because this is our destination

F: 4  $\Rightarrow$  shortest path from A to F is

**A  $\rightarrow$  D  $\rightarrow$  G  $\rightarrow$  F**

#### Problem 4:

4, BFS:

$Q = [A]$

$visited = \{A\}$

take A, add B, C, D

$Q = [B, C, D]$

$visited = \{A, B, C, D\}$

take B, add E, F

$Q = [C, D, E, F]$

$visited = \{A, B, C, D, E, F\}$

take C, no add,  $Q = [D, E, F]$

take D, no add,  $Q = [E, F]$

take E, no add,  $Q = [F]$

take F, no add,  $Q = \emptyset$

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F$

DFS:

$Stack = [A]$

pop A, push D, C, B,  $stack = [D, C, B]$

pop B, push F, E,  $stack = [D, C, F, E]$

pop E, no push,  $stack = [D, C, F]$

pop F, ———,  $stack = [D, C]$

pop C, ———,  $stack = [D]$

pop D, ———,  $stack = \emptyset$

so  $A \rightarrow B \rightarrow E \rightarrow F \rightarrow C \rightarrow D$

#### Problem 5:

Pseudocode:

rod-cutting-wCost(price, n, c)

create array  $dp[0..n]$

$dp[0] = 0$

```
for j = 1 to n
    best = price[j]
    for i = 1 to j - 1
        best = max(best, price[i] + dp[j - i] - c)
    dp[j] = best
return dp[n]
```

This problem is more complicated than the original problem which is just cutting rod, because when we cut now we have to think whether cost < revenue or not.

We set `best = price[j]` because maybe no cut is better than cut, because cut may make cost > revenue.